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The Twiss-Hanbury Brown Controversy : A 40-Years Perspective †

by

P. Goodman, W. S. Langer and S. P. Brumby,
School of Physics, University of Melbourne, Parkville, Vic. 3052,

' "You should learn to say what you mean" the March Hare went on. "I do -at least I mean what I say - that's the same thing you know!" Alice hastily replied. "Not the same thing a bit!" said the Hatter'. C.L.Dodgson, 1865.

ABSTRACT

Some key points in the debate over the visible light "intensity interferometer", initiated by Hanbury Brown and Twiss's 1956 observation of a positive correlation between two photomultiplier signals from a split-beam optical source, are re-examined. This was a preliminary to their first measurement of a stellar diameter by this means, made in the same year, in the face of great experimental difficulties and criticism of their method, which produced a value for the angular diameter of α-Sirius which is not only in agreement with present-day measurements, but which provided one of the first tests of quantum electrodynamical theory, and considerably stimulated its further development. We also examine the role which could still be played by "intensity interferometry", which would be complementary to the programs currently employing Michelson interferometry, by analogy with present-day high-resolution electron microscopy.

1.INTRODUCTION

The January 7th. 1956 edition of *Nature* carried a brief two-page article by R.Hanbury Brown and R.Q.Twiss entitled "*Correlation between photons in two coherent beams of light*" (Hanbury Brown and Twiss,1956a) reporting a theory and a laboratory test, which was to set in train a debate which has endured the passage of time. The two remarkable aspects of the proposal to employ intensity correlation to measure "coherence" were firstly that this idea was anathema to most physicist of the day, and the fact that the proposed method actually worked. Within the same year they published an experimental result of 6.8 mas (milli-arc-seconds) for the diameter of α-Sirius (Hanbury Brown and Twiss,1956b), in reasonable agreement with the bolometric value (6.3 mas) then available. It was in this article that the term "Intensity Interferometer" was first used.

The present article sets out to examine the nature of questions raised by these land-mark publications, and in the process to shed some light on the nature of prejudice itself, of the kind which traditionally leads physicists to doubt any radical departure from currently held beliefs. The controversy occurred in two periods, namely 1955 → 1964, (the *Nature* debate), and 1990 → 1994, as set out in an appendix, in Tables A1 and A2.

The use of 2-beam interferometry in astronomy was established by Michelson (1891) who used the

† This paper is dedicated to Professor Hanbury Brown on the occasion of his 80th. Birthday.

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technique to measure the diameters of four of Jupiter's satellites. Michelson and Pease (1921) and Pease (1925; 1930) then measured several stellar diameters between 1920 and 1930, initially with a 3 metre baseline, and succeeding with a moderate extension (<15 meters).

Hanbury Brown and Twiss came from the somewhat different background of British radar, which had led to the discovery of astronomical radio sources shortly after the war. These radar dishes were equipped with angular tracking systems, and sophisticated signal amplification electronics. They had successfully devised and operated a radio-signal correlator, using two radar dishes with a variable separation r_0 , focussed on the same astronomical source, measuring the angular fall-off of signal-peak correlation with r_0 . This involved sending both "fluctuating" signals through amplifiers to a central signal multiplier, which acted as a self-correlator, rejecting signal components not common to both sources. It was important to achieve a sufficient S/N (signal-to-noise) ratio, and for this purpose the 'noise' was judged to be the uncorrelated signal obtained from one detector in the same band-width (Hanbury Brown and Twiss, 1954).

The method worked well , but when they found that the radio sources were relatively large, they turned their attention to visible-light sources. This as it turned out was a bold step, for the simple reason that the detection systems for radio and visible light were (and are) radically different, but they were sufficiently confident of their physics to proceed. The content of the initial debate (1955 - 1964) therefore tended to be centred around the detector physics, which shifted later (1990-1994) when it was realised that emission physics rather than detector physics was central to the "bunching" effects being observed. Hanbury Brown and Twiss were motivated to trace their positive results in two-channel correlation as arising from photon and finally photoelectron correlation at the detector.

The basis of their ideas was the intuition that visible light sources were likely to be modulated over a very wide band-width, and that high-temperature stars in particular would emit a bunched, or highly fluctuating light train. They reasoned that such emission behaviour was basic to boson statistics for a point source with a finite wavelength spread, and proceeded to search for fluctuations in the 5MHz to 45 MHz range which were detected simultaneously by two matched and independent detection systems, using auto-correlation. They chose to integrate the correlated part of the signal, rather than to use coincident circuitry to directly correlate individual peaks, as advocated by others. They suggested that counting of "photons" (the latter approach) represented the particle rather than the wave approach, immediately introducing the notion that particle-wave duality for light, which was not resolved satisfactorily before the advent of quantum electrodynamics in 1950, was a key issue in the physics of photon bunching.

2. THE INITIAL DEBATE , 1956-1964: "PHOTONS AND COHERENCE".

The initial publication (Hanbury Brown and Twiss, 1956a) described a laboratory experiment in which a mercury discharge lamp was used to form a collimated beam split using a half-silvered mirror. Two photocells were then used as detectors and their integrated signals amplified and passed through a signal multiplier using the 3 - 27 MHz band. A positive correlation was detected using this "partially coherent" source, from which they concluded that "photons in two coherent beams are

correlated, and that this correlation is preserved in the process of photo-emission". Brannen and Fergusen (1956), who were strongly critical, concluded that there was "no correlation (less than 0.01%) between photons in coherent light" on the basis of results obtained from coincidence counting techniques, and added that if such a correlation did exist it would call for "*a major revision of some fundamental concepts in quantum mechanics*".

However two months after the Brannen and Fergusen paper, Hanbury Brown and Twiss had published their results a stellar "intensity interferometry" (Hanbury Brown and Twiss, 1956b). Their Table 1 from that paper gave a plot of observed correlation vs. base-line (4 points, for 2.5, 5.6, 7.6 and 9.2 metres respectively), giving the experimental curve which allowed them to make an initial estimate for the apparent diameter of α-Sirius. These measurements took 5 months to complete under the difficult conditions prevailing in Manchester, an argument for making further measurements from Australia instead of at Jodrell Bank (a second "intensity interferometer" was built at Narrabri during the 1950's). A more detailed analysis of these results given in 1958 (Hanbury Brown and Twiss, 1958b) reveals something of the human condition of the experimenters at tthe time. The need to obtain positive results as speedily as possible meant that every possible occasion between November 1955 and March 1956 was utilised, yielding only 18 hours of observation. Approximately half the time was blocked out by moonlight; also restrictions were severe because Sirius is approximately 20° above the horizon at that latitude. Observations with the mirrors as close as possible were run to evaluate the signal-to-noise (S/N) ratio, and 5.7 hours of observation were obtained by 1955's end. "Throughout January 1956 attempts were made to increase the number of observations with the mirrors in their initial positions, but no results were obtained because of cloud, and it was decided to use the remaining time to measure the correlation with longer baselines and thus to find an experimental value for the angular diameter of Sirius." All the significant observations published in November 1956 were obtained essentially during February of that year, although the remaining months were required for statistical analysis. This timetable shows that by the time their publication on a laboratory source reached press their astronomical observations were already underway, and that by the time of Brannen and Fergusen's,1956) article they had already obtained what they regarded as positive proof of correlation using an astronomical source.

The key idea behind the "intensity interferometer" was that of 'beating-down' the high visible-light frequencies, which they believed would occur between coherent light signals of slightly different wavelengths, to bring some signal within bandwidths accessible to contemporary electronics. The bandwidth chosen, of 5Mhz to 45 MHz, would correspond to $\Delta\lambda$ values far narrower than normal spectral-line widths, and closer to hyperfine structure dimensions. The belief that a point source at astronomical distances with a finite $\Delta\lambda$ spread would lead to 'cross-colour' interference between neighbouring λ values and lead to time-dependent modulations of signal intensity was attributed to a property of boson statistics, permitting more than one photon to occupy the same "phase-cell". They also thought it natural to assume that such modulations would have a certain angular coherence, and that neighbouring points within stellar disks would have independent fluctuations.

The 1955-1958 Experimental Set-up.

The experimental set-up is shown in Fig.1. The equipment consisted of two standard army searchlight mirrors, paraboloids of borosilicate glass 156 cm. diameter and of 65cm focal-length (1958b). These could focus the light from a star into an area about 8mm. in diameter. The signals were collected by photomultipliers and amplified in the band 5 to 45 MHz. The R.C.A. photomultipliers used were of the multi-alkalie type, their cathodes used Cs, K and Na metal deposited by vacuum decomposition. One arm of the "interferometer" contains a variable time delay, needed to obtain a coincidence in time-scale for the two signals (in general the paths will differ in effective length). If these signals had been pre-recorded they could have been scanned, one across the other, to obtain a maximum in the signal product (the auto-correlation procedure). As it was the "time-delay" was relatively crudely implemented and consisted of plugging in different lengths of coaxial cable to obtain the best results. This 'on-site' procedure replaced the method described for the laboratory-source experiment (1957b) for which the distances between source and the 2 detectors was arranged mechanically to be equal to approximately 1 cm in an overrall distance of 2.65m, i.e.to ~0.4%. Once this value had been established it was left constant, and the peak signal value recorded for discrete values of mirror separation 'd ' to obtain their $\Gamma^2(d)$ curve. Using this data, which is analogous to far-field diffraction data, being recorded in "angular" space, it is readily understood that by Fourier integration the real-space dimensions of the source function (in this case a cylindrical source equivalent to the 1-dimensionally resolved star diameter) will be determined.

Poor S/N values meant long periods were required for measurement. However once the method (of "stellar intensity interferometry", now referred to as S.I.I., while the Narribri version is referred to as N.S.I.I., as in Table 1) was established, the method could be refined with each successive advance in electronics and particular in computing science which had only just begun, though it was perhaps unfortunate that a later revision of their statistics (Hanbury Brown and Twiss, 1958b) caused them to modify their estimated diameter for Sirius by +5%, which in fact deviated their value from the presently accepted one (see Table 1).

In their (1958a) paper they summarised the advantages to be gained over the Michelson stellar interferometer. Firstly, by setting the low-frequency cut-off at 5 MHz they had successfully excluded the atmospheric scintillations effecting all other earth-bound observations including the Michelson interferometer. In addition, the S.I.I. has a potentially much greater resolving power than the Michelson interferometer. This potential lay in possible extensions (as at 1958) of base-lines to around 2 km, (the largest working Michelson interferometer at the time had a 20ft. baseline), and at the present time the technique would permit very much longer base-lines (at least 100's of kM).

They therefore predicted that the method would have its greatest use for stars "earlier than G", in particular for resolving "the nearest of the W-, O- and B- stars". (Table 2 gives the astronomical nomenclature used).

The first major paper (1957a), subtitled "the correlation between photons in coherent beams of radiation", did little to end the debate. The treatments are detailed but concentrate largely (as in all earlier papers) on aspects of detector physics rather than of considerations of emission at the source. In the next paper (1956b) they discuss the important distinction of requirements for the two experimental approaches of (a) multiplication of continuous signals, and (b) coincidence counting of "photons", and having shown earlier that the coincidence method would require far more care in experimental procedures than previously given (Hanbury Brown and Twiss, 1956c), gave a detailed analysis of this alternative method (Twiss and Little,1959) which could now be used on a laboratory source to give a positive result for correlation (this followed earlier but briefer reports of successful laboratory measurements by Rebka and Pound,1957, and by Twiss, Little and Hanbury Brown, 1957). However it was already clear that this method had no applications in astronomy, at least with equipment available at the time..

There is however a real interest in the different mechanisms by which radio-frequency and visible light detectors work. In the former case direct interaction occurs between the oscillating radiation field and the conduction-band electrons of the antenna, the inherent low energy threshold in this case gives a far more favourable signal-to-noise ratio than in the optical case, where the photoelectric effect is involved.

This period of debate ended with the publication of an article by the original protagonists, consisting of Jánossy and his co-workers (Farkas et al., 1964), in which the essential feature of "photon bunching" was accepted ("a real beam of light shows fluctuations of intensity caused by the interference of wavebands emitted by individual source atoms, which are independent of intensity and are of the order of 100%"). They also sought to measure the fluctuating contrast of interference fringes, and appear to have succeeded in obtaining a positive correlation of fluctuations between the two channels using coincidence circuitry. Although these authors considered the physics of the source (which had finally emerged as the main point of interest) rather than that of the detector, their method had little in common with the Hanbury Brown-Twiss technique requiring a multiplication of signals. This publication does however represent a major backdown on the fundamental objections based on quantum mechanical issues.

3.THE SECOND PERIOD OF DEBATE: 1990-1994: DIRAC'S DICTUM.

Dirac gave his now famous views in Chapter 1 of the 1st. edition of "*Quantum Mechanics*" (Dirac, 1930), in two statements: (a) "*Interference between two photons never occurs.*" . .for if this were to happen. . (b) "*Sometimes these two photons would have to annihilate one another and other times they would have to produce four photons. This would contradict the conservation of energy*". These statements were initially made before the development of quantum electrodynamics, but students of Dirac's work will see that although many sections of the book were revised by the time of its fourth edition (Dirac, 1958), this paragraph remained intact.

R.M.Silllito of Edinburgh University (Sillito, 1957) was the only author in those listed in Table A1 to make explicit the quantum electrodynamical arguments required, explaining why Dirac's statement (b) must be rejected in responding to the H.B-T experiments, as follows:

"(1). During the time that photon wave-packets overlap the state of the 2-photon system must be symmetrical in the 2 photons due to Bose statistics (the 2 are indistinguishable).

(2). This *does not imply that 2-photon interference may lead to 4 or no photons* . The photon is not a particle: it does not survive a counting process unchanged. Rather, the interference between photons produces a distortion of the time distribution of events by which photons are detected."

This points to the flaw in Dirac's justifying statement (b), i.e. it is the classical-particle theory of photons rather than photon interference itself (as assumed by Dirac) which fails to conserve energy. His statement can be understood from a 1930 perspective, its repetition in much later issues of his book led many physicists to treat it as a current theoretical truth.

Dirac's statement (a) on two-photon interference could be defended on different grounds prior to 1960. At that time, a coherent *optical* source had only one meaning: it was a point source giving rise to spherical-wave propagation. After 1960 and the invention of the laser this was no longer true, and the advent of coherent sources of finite cross-sections changed the demands on basic theory. With the development of phase-locked lasers it became possible to "simulate" Young's experiment with two lasers replacing the two slits which were the alternative paths for any single photon. The introduction of laser interference spectroscopy (Javan et al., 1962; Magyar and Mandel,1963) soon started fresh debates about the physics of the Hanbury Brown-Twiss interferometer, the most recent being that carried in the American Journal of Physics over the period 1990-1994. There were relatively few contributors here, but the weightiness of the matters discussed and the general lack of enthusiasm for criticising ideas from so eminent a physicist as Dirac can be judged by the lengthy period between receipt and publication by this journal (Table A2). The initial paper for example (Louradour et al. 1993) carries the editorial comment "received November 1990; accepted March 1992".

One of Louradour et al.'s experiments involved overlapping two picosecond pulse emissions of band-width-limited duration from two separated lasers and displaying the resulting interference pattern on a screen. They found fringes with close to 100% contrast, with both lasers emitting a central wavelength value of 5680 Å. A second experiment involved laser beams of *different* frequencies, with $\Delta\lambda = \lambda_1 - \lambda_2 = 0.25$Å, to obtain difference-frequency beats, a phenomenon they described as "spatio-temporal interference between two beams of ... different frequencies". They also recorded the beat contrast as a function of beat frequency, which they could vary using distributed-feedback dye-tuned lasers (DFDL's). This type of experiment has since become a subject in itself, closely associated with establishing better standards for units of time and length.

Louradour et al., referring firstly to Magyar and Mandel's (1963) earlier work as well as their own, claimed that these results finally showed Dirac's dictum to be invalid. In response, Wallace (1994: "received April 1993; accepted July 1994") commented: "a single photon may .. originate in separate lasers. There is no flaw in Dirac's argument, *which should not surprise anyone*. "

The finally paper of this series (Davis and Parriger, 1994: "received May 1993; accepted July 1994"), attempted a reconciliation by supporting neither side directly, stating:

(1). " ...Pflegor and Mandel's (1968) fringes (between two lasers) were recorded even when single photon detection was used, and with laser beams so severely attenuated that 'with *high probability* one photon is absorbed before the next one is emitted'.. demonstrating that *first order interference* is a single photon effect in support of Dirac (1930-67). *Dirac's explanation ... can be regarded as the centerpoint of quantum theory*".

(2). " since a photon is a quantum mechanical particle its localisation within an interference pattern results in our inability to ascribe it to either of the sources separately".

(3). " since the frequency difference between the two beams is greater than the band-width of each beam, each detected photon is 'bichromatic': its localisation within an interference fringe results in our inability to ascribe it to either (laser) separately".

(4). while Dirac's statement is correct for first order Young's interference effects, interference between two different photons *does* occur in higher order correlation experiments as first demonstrated by Hanbury Brown and Twiss (1956b).

This paper serves to highlight the dilemma facing some physicists in attempting to reconcile the irreconcilable. It is (not surprisingly) internally inconsistent (how can the "centre-point of quantum theory" be inapplicable in higher-order experiments?), while statement (3) would appear to be in conflict with accepted quantum electrodynamical definitions.

In this way the results obtained by Hanbury Brown and Twiss for their S.I.I. had much more far-reaching effects on physics, in the form of providing a proof for the predictions of quantum electrodynamics, than they could have ever anticipated or even desired.

4. MOTIVATIONS FOR CHANGE.

While Sillito (1957) was clearly the first to appreciate the significance of the Hanbury Brown-Twiss results in supplying experimental evidence for quantum electrodynamical theory, to gain a proper historical perspective one needs to look at other more deliberate attempts to investigate multiple-photon interference which had taken place, both prior to and after the introduction of laser interferometry.

In a theoretical treatment, Dicke (1954) investigated the consequences of spontaneous emission by two initially-excited atoms, obtaining cross-product terms equivalent to those obtained much later, the non-classical nature of his result being stated explicitly in his second paper (Dicke,1964).

Then, Forrestor et al.(1955) successfully produced beats between two Zeeman components from the ^{202}Hg line at 5461Å , overcoming severe S/N problems using a frequency-chopping technique. The first demonstration of the generation of beats from two lasers was given much later (Javan et al. 1962). A description of Forrester et al.'s experiment and of the Dicke (1954,1964) theory has been given by Paul (1986).

Hanbury Brown and Twiss, although aware of current theoretical prejudices with respect to visible light, approached their problem from a simpler and in some ways more fundamental basis.

Since interference between two lower-frequency radio sources was an everyday experience, they argued that the prejudice that visible light was "different" arose from the difference in detector systems, and not from any basic physics. It is clear that they had no wish to challenge Dirac's statement, which they supported in their publications, and which they were (unwittingly) about to disprove.

5. DEFINITIONS AND LANGUAGE: THE IMPLICATIONS OF FIELD QUANTISATION.

The contemporary definition of a "photon" dates from Einstein (1905) with his proposal that continuous electromagnetic radiation of wavelength λ be regarded as a collection of *massless* particles of energy $E = h\omega$, which built on Planck's (1901) proposal for the quantization of radiation. The zero-mass property alone ensured that a photon could never be treated as a classical Newtonian particle. The accepted definition of a photon is therefore related to the transition between atomic states (ground state and excited state for example), a transition which involves one bound-state electron, and the emission (or absorption) of a photon of energy, $h\omega = E_2 - E_1$.

For mathematical convenience we assume energy states are "clean", whereas in reality the states are "smeared" with energy values in a range ΔE dependent on temperature and other factors. We define a photon as a specific mode of the radiation field with a single $\boldsymbol{k}$-vector and polarisation state, while also referring to single-photon detection (via the photoelectric effect) on a time-dependent basis for which there must be a finite wavelength spread, following the Heisenberg relationship $\Delta t.\Delta E \cong h$. A further time-sharpening process is invoked when our detection system is followed by a high-frequency pulse-detection amplifier which behaves as a narrow band-pass filter.

In classical electrodynamics radiation fields are generated by vector potentials satisfying certain gauge-fixing conditions such as the transversality condition, $\nabla \cdot \mathbf{A} = 0$. The fields are obtained via the usual Maxwell equations,

$$\mathbf{B} = \nabla \mathrm{x}\, \mathbf{A}, \qquad \mathbf{E} = -\frac{1}{c}\frac{\partial \mathbf{A}}{\partial t} . \tag{1}$$

Treating the atom as a dipole oscillator interacting with an external, classical E.M. field provides a framework for discussing scattering. For this purpose we first make a Fourier expansion for the time dependent vector potential $\mathbf{A}(x,t)$ in momentum space (since our observations are at asymptotic infinity, the regime of far-field diffraction), generating time-dependent Fourier coefficients

$$C_{k,\alpha}(t) = C_{\mathbf{k},\alpha}(0)\exp(-i\omega t), \text{ and } C^*_{k,\alpha}(t) = C^*_{\mathbf{k},\alpha}(0)\exp(i\omega t), \tag{2}$$

for wave-vectors $\mathbf{k}$, and polarisation states α. These latter may be circular or linear, to suit a particular experiment. For further discussion we will abbreviate the suffix $(\mathbf{k},\alpha)$ to (i), for convenience.

Second quantization is introduced by a re-interpretation of the potential $\mathbf{A}(x,t)$ as an operator acting on the photon states of the radiation field, so that,

$$C_i(t) \rightarrow c\sqrt{h/2\omega}\, a_i \text{ , and } \quad C^*_i(t) \rightarrow c\sqrt{h/2\omega}\, a^\dagger_i \text{ , } \tag{3}$$

where a_i and $a^\dagger_i$ are the annihilation and creation operators.

The operators $a^\dagger$, a are non-Hermitian but their product, the number operator,

$$N_i = a^\dagger_i a_i \ , \qquad (4)$$

is Hermitian, and may be represented by a diagonal matrix with integer elements from which the negative regime is normally excluded, since the product $a^\dagger a$ is a positive definite operator and negative occupation states are excluded. The eigenvector equation,

$$N_i|n_i\rangle = n_i|n_i\rangle \ , \qquad (5)$$

defines eigen values n_i as the occupation numbers for a specific $(\mathbf{k},\alpha)$ state, and represents the number of photons present in that particular state, while $N = \sum_{\mathbf{k},\alpha} N_{\mathbf{k},\alpha}$ is the operator measuring the total number of particles.

The action of $a^\dagger$ and a can be seen from the action of each on the vacuum state, represented by $|0\rangle$, the state in which no photons are present at a time t = 0, as

$$a^\dagger_i|0\rangle = |1\rangle, \ a_i|0\rangle = 0, \qquad (6)$$

i.e. $a^\dagger$ acts to introduce, and a to remove a photon from the system. The 0 is not the vacuum again, but the zero vector of the Hilbert space (since negative occupation has been ruled out, see above).

The introduction of time development shows that for each case this is reversed, with

$$a^\dagger_i(t) = a^\dagger_i(0)\exp(i\omega t), \ \ a_i(t) = a_i(0)\exp(-i\omega t). \qquad (7)$$

We next want to understand what is meant by a "semi-classical" treatment. We will adopt the working rule that where an experiment (or treatment) has a classical analogue it is amenable to semi-classical treatment, which is to say that at some parameter limit the quantum-mechanical and classical treatments are equivalent. In the optical case of present interest, where we have a distant source and an earth-bound detector, both source and detector should be treated quantum-mechanically, but transmission of the E.M. field through space can be understood from a classical wave (or particle) equation. This semi-classical treatment is not always applicable, in which case we have an essentially quantum-mechanical phenomenon. With fermions such an example would be superconductivity, which we note occurs in a situation with a *high* volume density of electrons (in a solid).

Returning to the present problem, a star is regarded as a source of thermal photons, while the detector is a photocathode. The quantum-mechanical probabilities, in general form, for both these processes, are,

$$\mathbf{A}^{(\text{emis})} = -\frac{e}{m}\sqrt{\frac{(n_{(\mathbf{k},\alpha)}+1)\hbar}{2\omega V}}\ \langle B|H_{\text{int}}|A\rangle \exp(i\omega t) \qquad (8a),$$

$$\mathbf{A}^{(\text{abs})} = -\frac{e}{m}\sqrt{\frac{n_{(\mathbf{k},\alpha)}\hbar}{2\omega V}}\ \langle B|H_{\text{int}}|A\rangle \exp(-i\omega t) \qquad (8b),$$

in which A and B represent initial and final states for the atom involved in the interaction, so that the "cross-sections", or intensities obtained from $|A|^2$ will be:

$$\sigma^{(\text{emis})} = c\,\frac{e^2}{m^2}\,\frac{\hbar}{2\omega V}\,(n_i + 1)\,, \qquad (9a)\,,$$

$$\sigma^{(\text{abs})} = c\,\frac{e^2}{m^2}\,\frac{\hbar}{2\omega V}\,n_i\,, \qquad (9b)\,,$$

with the factors n and n+1 arising from the $a^{\dagger}$ and a operators. In going from eqns. (8) to (9) we have assumed the approximate solutions given for the interaction term to be solved with suitable approximations for the cases of thermal emission and photoelectric absorption: their solution is given in several texts (Sakurai, 1967; Mandel and Wolff, 1995) and contribute only parametric constants to the equations.

These compare with the equivalent classical expressions, where we have the energy density of the E.M. wave, $(\mathbf{E}^2)_{ave} = \bar{n}hc/\lambda)$, with $\bar{n}$ the photon density per unit volume,

$$\sigma^{(emis)} = c[\bar{n}\hbar(c/\lambda)] \qquad (10a),$$

$$\sigma^{(abs)} = c[\bar{n}\hbar(c/\lambda)] \qquad (10b).$$

Comparing eqns. (8) and (9) we note that, unlike the case for fermions, the classical expressions are approached for *high* values of n, when n and n +1 become indistinguishable, but deviate most strongly *for the emission case only*, for small n, i.e. for very weak beams.

This is important in reviewing the Twiss-Hanbury Brown debate. These authors, in both (1956a) and (1957a) papers, went to great pains to show that their photo-multiplier detectors would act as square-law detectors, and in their analyses give a detailed analysis of this situation in terms of photoelectron rates, even though their signals were integrated over time. The same can be said for other authors (Purcell, 1956; Felgett 1956) from Table A1. From the above we can see that the interest lies in the source rather than the detector, and that their conclusion on detector response is expected from theory.

Another claim, that "diluting" the source (by distance as in the case of a star) by reducing the "chance of overlap between photons in transit due to their sparseness" would reduce the bunching effect is also erroneous if the bunching occurs at the source. We may consider each such "point" on a stellar disk as a volume given by $\Delta x,\Delta y,\Delta\zeta$, where the area $\Delta x,\Delta y$, is beyond the resolution of the interferometer and an order of magnitude smaller than the total stellar area, and $\Delta\zeta$ is taken as parallel to the propagation direction, and generally $\Delta\zeta >> \Delta x$. The emission from stars in the age range, say, O to K, (Table 2) from such a volume is quite complex: spectral lines originating deep inside the stellar atmosphere, with very high atomic densities must propagate through the cooler and less dense atmosphere for considerable distances and through a range of temperature gradients. This results in a complex mixture of emission and absorption lines, and this spectra as recorded shows a range of fine detail which goes beyond the resolution of currently available spectrometers (see Fig.2). This situation is ideal for the type of experiment envisaged by Hanbury Brown and Twiss in 1955. Rough calculations show for example that for λ = 5550Å, Sirius produces approximately 10^{-16} Watts/cm^2 at the earth's surface, which amounts to a photon collection frequency of around 3 x 10^4 /sec., or 30 KHz, a factor almost 7 x 10^{-3} down on the cut-off frequency of 5 MHz used in the S.I.I. experiment. Spatially, this gives a photon separation of around 10 km, an order-of-magnitude greater than the coherence length for stellar sources. The band-width used in 1956 of 5- 45MHz. ($\tau \approx 2\text{x}10^{-7}$

to $2x10^{-8}$ sec) corresponds to wavelengths in the region of say 6 - 60 meters, resulting from photon interference.

In conclusion it is worth recording that the first successful time-resolved experiments with thermal light confirming bunching effects, were not reported until the late 1960's. This was due to the requirement for faster-response photo-cathodes, and faster response circuitry, than that available in 1956. This was carried out by Morgan and Mandel (1966) using a ^{198}Hg isotope discharge source, obtaining a spectral line coherence of ~ 2ns which compares favourably with the then-available photo-response of ~ 1ns. Using coincidence circuitry they were able to show beyond reasonable doubt that such photons tend to be emitted in bunches rather than at random from a thermal source.

6. ANTI-BUNCHING EFFECTS WITH PHOTONS.

Immediately after the bunching effect in photons was understood, several proposals were put forward by different laboratories to begin looking for anti-bunching effects using a beam of fermions, and neutrons in particular were considered because of the absence of coulombic repulsion. However no such experiments were ever carried out because as we have seen, very intense beams would be required for fermions, well beyond any experimental laboratory source. However by the 1970's anti-bunching for photons had been observed, though this cannot be done for either laser or thermal sources for which, from earlier discussion, semi-classical arguments can be applied. What is required is a source which is purely quantum-mechanical; such a source may be obtained by the phenomenon known as "resonant fluorescence", which occurs when an atom is held in a magnetic field, causing Zeeman splitting of the spectral lines, and requires the use of two lasers. Laser 1 is used to generate the emission lines from the Zeeman states to the ground state, emitting photons which are now circularly polarised. Then a second laser is used to keep the atom in continual Rabbi oscillation, between ground and excited state, giving the time-averaged value for the emission energy of $\frac{\hbar\omega}{2}$. This introduction of the factor 1/2 is all that is needed to change the statistics to anti-bunching, and interestingly enough involves now the negative sector of the number matrix, which we earlier discarded for all "normal" sources. This phenomenon was first observed by Kimbel, Dagenais and Mandel (1977, 1978), and later by Dagenais and Mandel (1978).

7. A FUTURE FOR COMBINING MICHELSON AND INTENSITY STELLAR INTERFEROMETRY USING CONTEMPORARY ELECTRON CRYSTALLOGRAPHY TECHNIQUES?

It is a matter of considerable disappointment to many of those involved, that S.I.I. was discontinued soon after the N.S.I.I. measurements in 1974 (Hanbury Brown, Davis and Allen, 1974). In the meantime however Michelson stellar interferometers have been advancing very greatly, and 2 or 3 designs for space-station observatories have appeared which exploit 2-dimensional arrays of relatively small-aperture telescopes. In these, 2-dimensional optical images of the stellar source are obtained, by combining the individual 2-dimensional telescope images pixel by pixel, within a digital computer. For optimal results the optical path for all telescopes should be zero, but where this is not a

practical goal, tests are run to estimate the maximum mean phase delay between inputs which will lead to a tolerable degree of "defocus".

It has occurred to us that this system has a close analogy to present-day HREM (or high-resolution electron microscopy), where an objective aperture replaces the individual telescope apertures in the above type of interferometer, and the symmetrical array of aperture-limited diffracted beams seen when a single crystal is aligned to a zone-axis of particular symmetry. Both systems allow for example the true symmetry of the source/crystal to be imaged, albeit at limited resolution.

This gives quite different information from S.I.I. which relies on the auto-correlation function,

$$f(r_1)g(x+r_1)dr_1 \ ,$$

when $f(r_1) = g(r_1)$,

which automatically guarantees that asymmetric objects will acquire a centre-of-symmetry, for example, in "imaging" a comet (by Fourier-transforming the intensity data), the comet would appear to have 2 tails, symmetrically disposed about the centre. This is a direct consequence of Friedel's law in kinematic diffraction (1st. Born approximation scattering), common to x-ray diffraction and biological-material electron diffraction (Dorset, 1995), as well as to simple optical diffraction, where for example a triangular slit when narrowed will give rise to an hexagonal star in the Fraunhofer plane.

We are not by any means the first to show this parallel of astronomical source resolution and crystal diffraction resolution, i.e. that both these phenomena occur in the Fraunhofer plane. In a paper perhaps now largely overlooked, Goldberger et al. (1963) pointed this out in great detail, and even described how the S.I.I. correlation method could be applied to x-ray diffraction in order to solve the so-called "phase problem" arising from Friedel's law.

Recently however a new field of biological crystallography has been opened up, mainly by Sass and co-workers and Jap and co-workers, in studying bacterial membranes known collectively as Porins, which involves combining the phase-retaining information of limited aperture from HREM with that of the intensity data collected over a much greater aperture by electron diffraction. This is done in an iterative manner (Sass et al.,1989; Walian & Jap, 1990), until finally a high-resolution image with the correct phasing details (a "reconstructed image") results. Without going into procedural details we present in Fig.3 (a) a projected image of the structure of Omp F porin, obtained and processed by A. Massalski (Dorset, 1995), showing a perfect trigonal (i.e. non-centrosymmetric) symmetry, at a resolution of ~ 6 Å. The in Fig.3 (b), from the later work of Walian and Jap (1990) from the closely related compound PhoE Porin, we see the zone-axis diffraction data, which is recorded over an aperture of ~ 2.8 $Å^{-1}$ as *intensities*, of the form $\psi\psi^*$.

Fig. 3 (c) then shows a reconstructed image, utilising phase data from a direct image (or several such images to increase S/N), combined with weighting factors from the wide-aperture diffraction pattern intensities. (Jap et al., 1990). This for the first time shows the biologically-significant departure from exact trigonal symmetry, by fine detail, particularly in the regions between

trimer structures. The resolution achieved here is estimated as ~ 3.5 Å, needed to resolve these features.

How could such algorithms be applied in astronomy? To understand this we need first to look at some new projections for stellar astronomy, based on two interesting designs, one intended for space-launch and the other designed for ground operation. The first of these is the MUST (Multi-mirror Ultraviolet Solar Telescope) interferometer whose plan is shown in Fig. 4 (a) (L. Damé, 1994). This consists of an array of five 20cm-aperture telescopes, giving an overall effective aperture of 71 cm, arranged pentagonally around a central recombination (non-collecting) telescope, which has been designed compactly for space launch within the European Space Space Station program. Its capabilities have been tested so far using simulated images of some solar-flare. Fig.4 (b) shows results obtained in this way of an asymmetric flare, with increasing deviations from the ideal condition of zero relative-phase-delay being entered into the simulation. This shows that an average phase delay for the telescope set of around 20° should still allow a satisfactorily sharp images to be obtained.

The compactness and robustness of this construction, and the use of small 20cm. aperture telescopes which can be made with near-perfect mirrors, free from the surface ripples caused by polishing a larger mirror, but which is also much less bulky and delicate than the large single-mirror design, make it suitable for satellite launch. However its overall effective resolution is limited to an aperture of 71cm. While this is not an issue for observations on the sun, when a more distant source such as Sirius is considered, it means of course that such surface details could never be observed.

A second interesting development is that of the CHARA (The Centre for High Angular Resolution Astronomy) array, shown in plan in Fig.4, designed for ground operation, and location on top of a large mesa in New Mexico. This array is an optical and infra-red imaging array of 7 x 1-m aperture telescopes in a Y configuration contained within a 400-m diameter circle, designed to analyse young "stellar objects", operating in the 0.55–0.9μm and 2.1–2.5μm wavebands (McAlister, Bagnuolo, ten Brummelaar, Hartkopf and Turner, 1994). Four fixed telescopes in this array include one central one, and a triangular surrounding array; these together with the three outermost and mobile telescopes provide the maximum r_0 (detector separation) value, based on the circular aperture, of 400m.

On the other hand, with the current advances in communication technology, earth-bound S.I.I. detectors could be used with much greater separations, i.e. of several 100km, relying on telecommunication dishes to transmit their signals to a central receiver. Of course, to compare with the electron crystallographic example earlier one would need a whole array of intensity detectors, but one has to bear in mind that here we are observing a single source; in the crystallographic case this would correspond to observing a single asymmetric molecule rather than an infinite periodic array, in which case the Fraunhofer pattern would be continuous rather than an assemblage of δ-functions, although would still be 2-dimensional, and the mathematics is not changed (simply a summation is replaced by an integral). However there are two courses of action we should consider here which

would allow S.I.I. to have a future role in resolving astronomical sources. These are (1) we plan to utilize the technique to enhance the phased resolution expected from the multiple-telescope-array Michelson-type interferometers at present being designed, or (2) we aim to obtain resolutions quite outside any available with current multiple-telescope-array Michelson interferometry.

For the first aim we would need to consider some 2-dimensional array, say a simple cross, with the S.I.I. dishes mounted on rails so that data could be obtained from r_0 values ranging continuously from say 200m to 2km for example, thus scanning the resolution region predicted for the CHARA array and going well beyond this limit.

Perhaps however it is more exciting (and funds could be more readily obtained) for planning a far more extensive base-line, making use of both space and communications technology currently available. One can then envisage a situation where two S.I.I. collector telescopes are placed in orbit, from two satellite launches, with an r_0 value of the order of say interplanetary spacing, transmitting to a collector satellite in stationary orbit which is in direct communication with an earth-bound receiving observatory. These signals could be time-correlated, since each observational station would be equipped with an atomic clock giving a recognisable train of timing signal pulses. Then, the principle of S.I.I. allows a collection of meaningful 2-dimensional observations to be made over a period of time τ, by orbiting the collectors within the observation plane, so long as τ is small compared with the time rate of changes in the source which may be of interest. Of course, with greater sophistication and expense, the dimensionality problem could be solved by launching 4 detectors in cross formation.

The value of such an observatory system should be in its ability to probe very early but bright stars, say, in the *W* and *O* class, and to probe these in detail using an optical spectrometer to separate different thermal features. Perhaps it is true that only now is the astrophysical theory sufficiently advanced where such information would be very useful in understanding stellar origins.

However our main task has not been to describe such an experiment in any detail; such a discussion must be left to others more competent in this field. Certainly the current author's would feel this whole article, intended as a tribute to the tenacity of Hanbury Brown and Twiss, would have been well worth the effort if it lead to the contemplation by the astronomy community of entering into such a program.

8. CONCLUSIONS.

In concluding this re-investigation of the controversy, we would assert that a large amount of this debate arose through a lack of mutual understanding of what was being discussed, which allows us to close with our opening reference to Dodgson (1985) as quoted at the head of the article, who to our knowledge was the first mathematician to insist on the importance of language as a prerequisite to meaningful debate (see also Wittgenstein, 1922). Regarding the influence, *in absentia*, of Dirac, his influential statements were made in reference to the quantization of the Dirac field rather than of the E.M. radiation field of concern in QED theory.

To end on a more constructive note, we believe that the S.I.I. could be re-employed with advantage using current technology to enhance our knowledge of distant stellar objects in finer detail, particularly if this technique could be combined with spectroscopy to separately resolve different emission-wavelength details.

ACKNOWLEDGMENTS.

The authors gratefully acknowledge assistance from Rachel Webster of our department in pointing out the type of problems which could benefit from new-technology S.I.I, from Dr. A.E. Spargo, Paul Dyson, Dr. Peter Coleman and Dr. Peter Tulloch for correcting and adding to our information on protein crystallography, from Dr. Andrew Booth of the Department of Physics, Sydney University, for alerting us to the existence of the Hawaii Conference Proceedings (SPIE, 1994), which proved a valuable source for contemporary stellar-interferometry technology. Since it is difficult to single out other individuals who have assisted, we more fairly wish to thank the Research staff and dedicated Librarian Kamala Lekamge of the School of Physics for their encouragement and material support during the course of this project.

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Table 1.

Estimates of the apparant angular diameter of Sirius α:1957-1994.

Publication year	Uniform-Disk Angular Diameter (mas)	Reference	Instrument
1957	6.3 ± 0.6	Hanbury Brown & Twiss	SII Mk.1
1958	7.1	Hanbury Brown & Twiss	SII Mk.1
1974	5.60 ± 0.15	Hanbury Brown et al.	NSII
1986	5.63 ± 0.08	J. Davis	SUSI Mk.1
1994	5.74 ± 0.08	J.Davis et al.	SUSI Mk.1I

Table 2.

Star Classification System.

W - Wolf-Rayet stars: Very hot and luminous, T ~ 100,000K.; less massive than might be expected from their luminosities which average ~ 10,000 times that of the sun.
O - Blue-white, T ~ 20,000-35,000K. Spectrum from multiply ionized atoms, especially He. Few absorption lines
B - Blue-white, T ~ 15,000K. HeII lines disappear, HeI appears in range B2 to B9. Neutral He, ionised O and Si and Mg; H lines stronger
A - White, T ~ 9000K. HI strong at A0 and dominate spectra; lines of ionized metals; some neutral metals.
F - Yellow-white, T ~ 7000K. H lines weaker than in A; lines of singly ionized and neutral metals.
G - Yellow stars like the sun, T ~ 5500K. Lines of ionnized Ca prominent; many ionized and neutral metals; H lines weak; conspicuous bands of OH radicals.
K - Orange-yellow, T ~ 4000K. Neutral metal lines predominate; OH bands.
M - Red, T ~ 3000K. ***C*** - Carbon stars, previously known as ***R*** and ***N***. Neutral metals; molecular bands, especially TiO_2.
S - Red low-temperature stars, T ~ 3000K.
Particular Examples: Sun (G2); Sirius A (A1); Betelgeuse (M2).

Table 3.

Comparative Data on Sun, Sirius A and Betelgeuse.

	Distance from Earth	Stellar Diameter	Parallax	Luminosity	Class
SUN:	149.6×10^6km.	1.4^6 km.	32′	-26.8	G2
SIRIUS:	31×10^{16}km.	1.07^6 km.	.0071″	-1.5	A1
BETELGUESE:	1878×10^{16} km.	$\cong 140^6$ km.	1.562"	-	M2

FIGURE CAPTIONS

FIG.1

Experimental set-up for the S.I.I. experiment carried out by Hanbury Brown and Twiss in 1956, in order to measure the apparent diameter of Sirius, showing collecting mirrors M_1 and M_2, the two photomultipliers D_1 , D_2 , with corresponding delay lines amplifiers, and a central signal-multiplication amplifier leading to both an integrater and, through a rectifier, to an RMS counter.

FIG.2

Section of a typical stellar spectrum, showing heavy absorption lines caused by transmission through the less-dense stellar atmosphere of the core radiation taken at the Crimean Observatory.

(a) Photographic record .

(b) the corresponding microdensitometer trace.

(Photographs from the book "Fundamentals of Astronomy" 22nd. Ed., Courtesy of Springer - Verlag, Berlin).

FIG.3(a)

Image obtained using a liquid-helium cooled cryomicroscope of the projected structure of Omp F porin, at ~ 6.5 Å resolution, showing an image with precise trigonal symmetry. The image was obtained and processed by Dr. A. Massalski. (Courtesy Plenum Press, N.Y.).

FIG.3(b)

Diffraction pattern from PhoE Porin extending to 2.8 $Å^{-1}$, obtained at -120°C using a JEOL 100B electron microscope fitted with a cryostage.(Wallan and Jap, 1990). (Photograph Courtesy of Academic Press, London).

FIG.3(c)

Projection grey-scale image of PhoE porin at an improved resolution of ~ 3.5Å which incorporates data from the Fig.6 diagram. (Photograph Courtesy of Academic Press, London).

.FIG.4.

(a) Telescope arrangement of MUST: 5 telescopes in a rectangular pentagonal array arranged around a central recombination station.(b) Series of images obtained by simulation, showing effects of focus defect in terms of the average departure of all 5 signal paths from zero phase difference. In sequence the top row shows phase departures of 0, 10°, and 20°, while the bottom row shows departures of 30°, 60° and 120°. This simulation is intended to show that an error as large as 20° will still give a satisfactorily sharp image, and only errors as large as 60° and 120° are too blurred to be useful. (Figures, Courtesy of James B. Beckinridge, Ed., SPIE, 1994).

FIG.5.

Telescope array for CHARA, showing a triangular array of telescopes, 4 fixed (black dots) and 6 movable (hollow circles).(Figure Courtesy of James B. Beckinridge, Ed., SPIE, 1994).

The total diameter of the outer circle here is 400m.

APPENDIX.

TABLE A1.

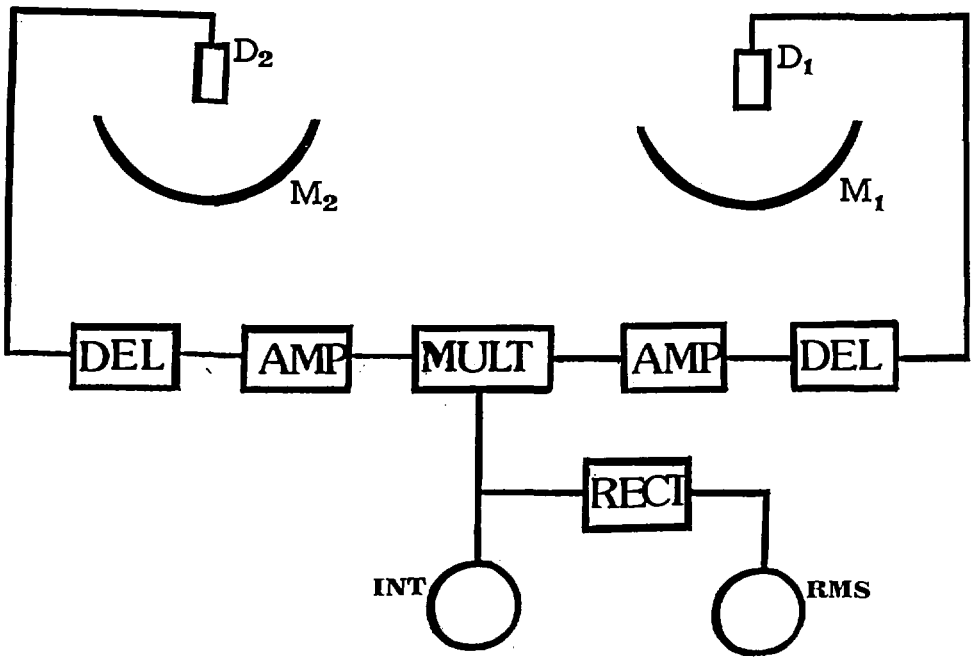
The Initial Debate: 1955 - 1964.`(Carried mainly in *Nature* = N in the Table).

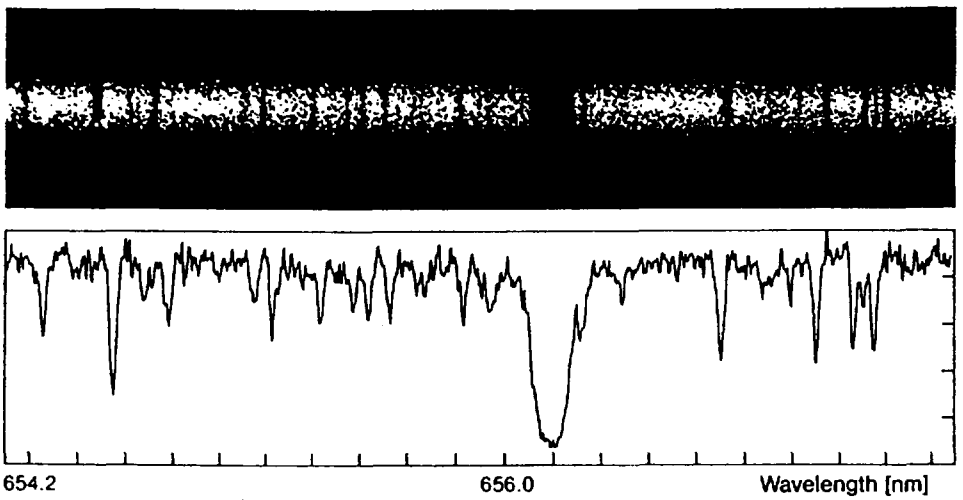
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TABLE A2.

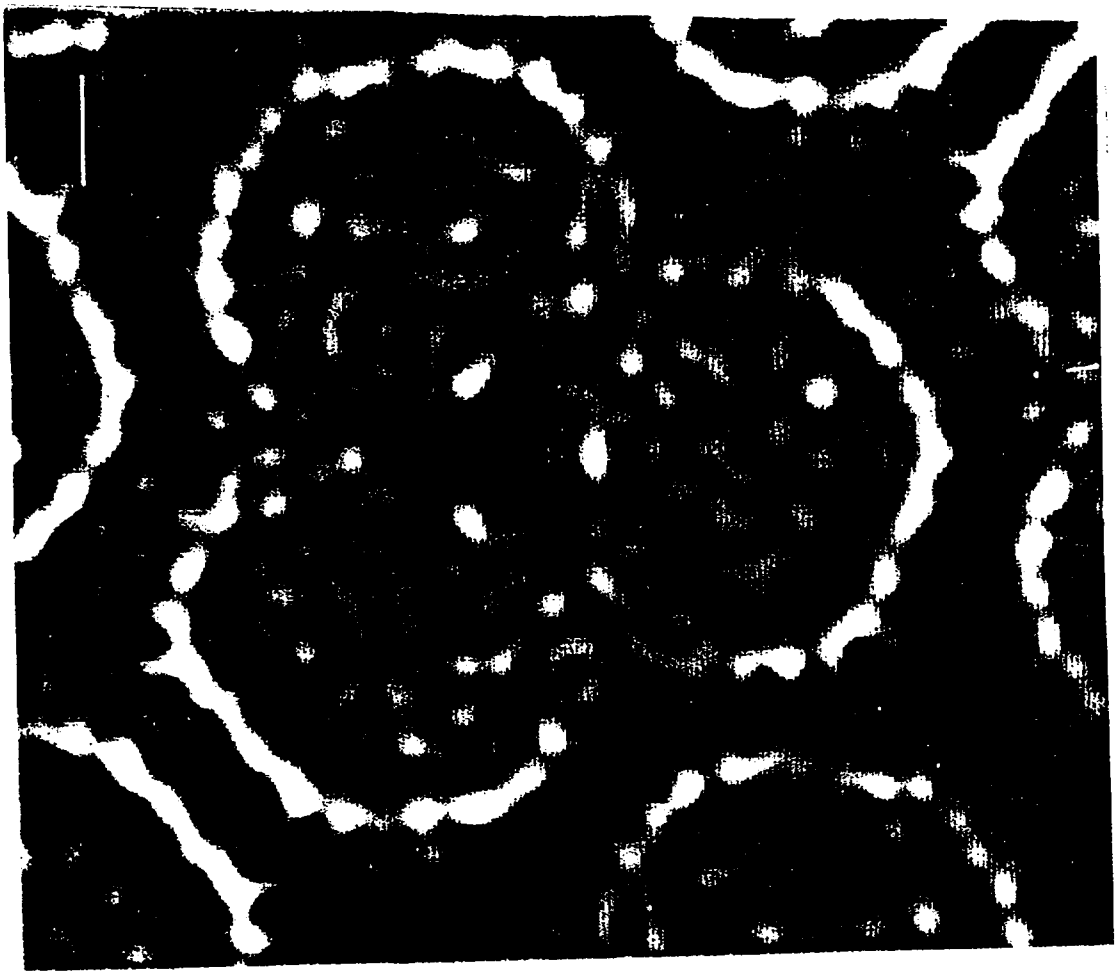
The Second Debate: 1990 - 1994.(conducted in the *American Journal of Physics)*.

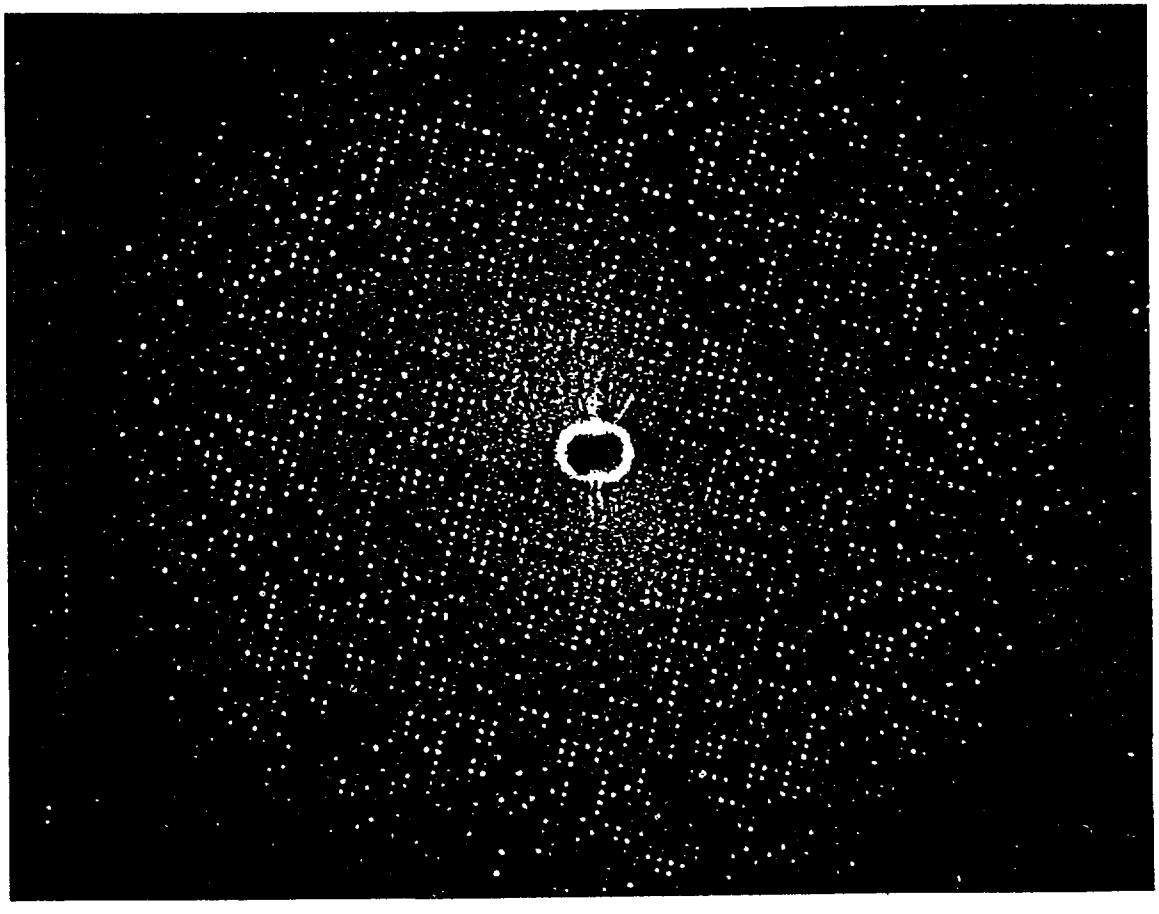
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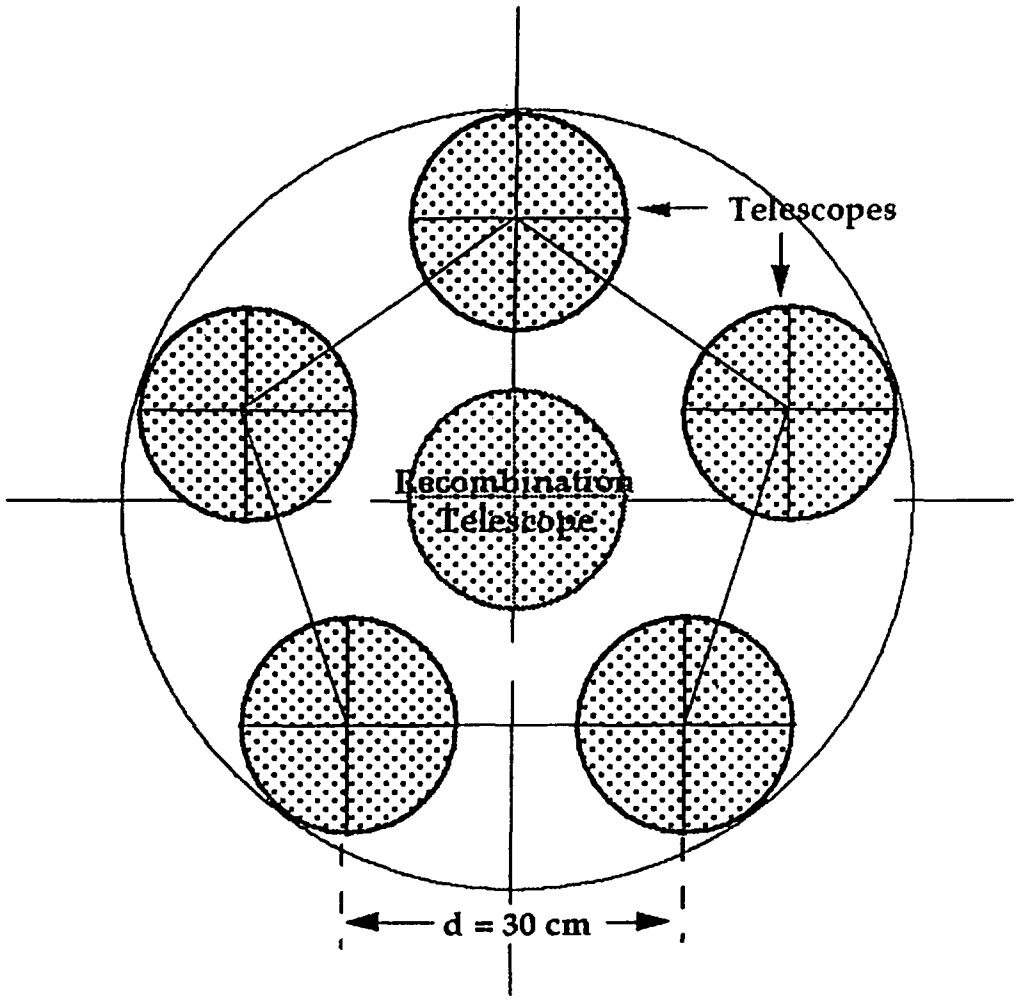

D2
D1
M2
M1
DEL
AMP
MULT
AMP
DEL
RECT
INT
RMS

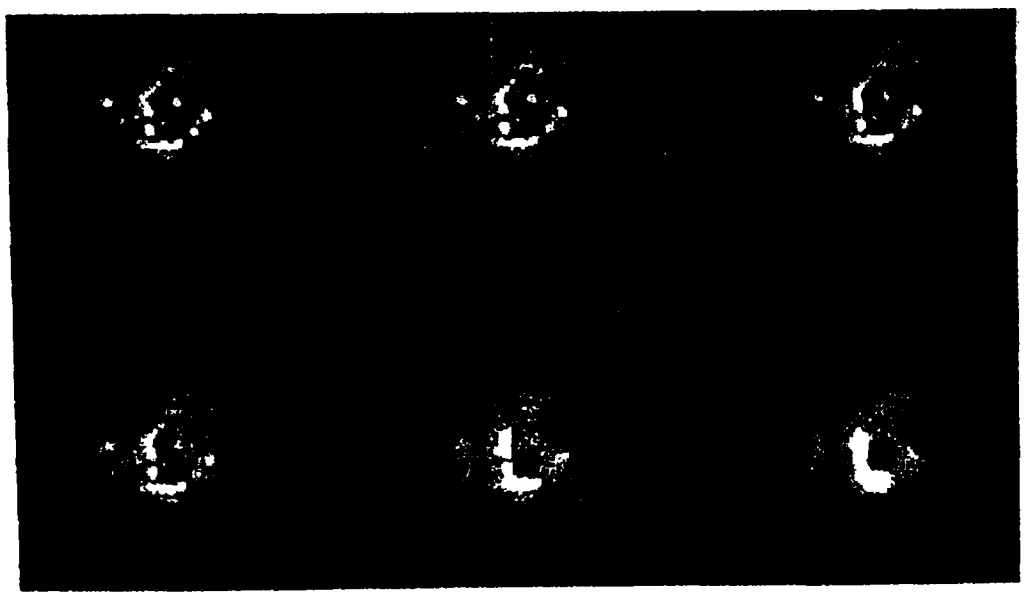

654.2
656.0
Wavelength [nm]

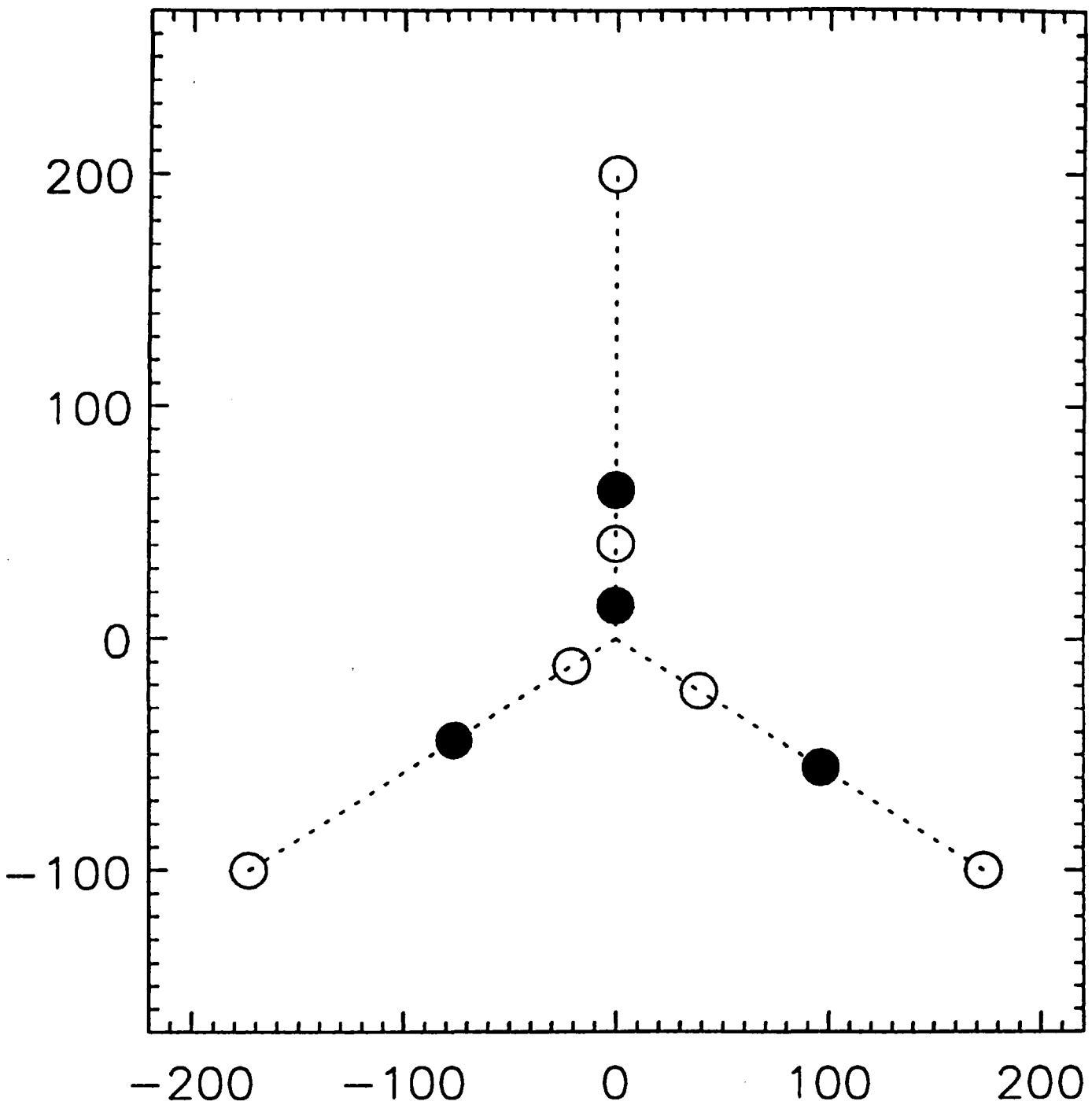







Telescopes
Recombination
Telescope
d = 30 cm




200
100
0
−100
−200
−100
0
100
200